INSTALLATION, OPERATING AND MAINTENANCE INSTRUCTIONS

SPENCE ENGINEERING COMPANY, INC.
Walden, New York 12586

TYPE ED SERIES PRESSURE REGULATORS

A. PLANNING THE INSTALLATION

1. Locate the valve in a straight run of horizontal pipe. See Fig. 1.
2. Allow headroom above the valve for access through the blind flange.
   Provide clearance for stem withdrawal underneath.
3. Prevent water hammer and erratic operation by installing traps to provide proper drainage before and after the valve, and before secondary PRV or control valve.
4. Avoid damaging effects of scale and dirt in pipe lines by using a SPENCE strainer as shown in Fig. 1.
5. Provide a 3-valve by-pass to facilitate inspection without interrupting service.
6. To eliminate excessive noise and erratic regulation with steam and other compressible fluids enlarge the delivery pipe size to effect a reasonable flow velocity at the reduced pressure. A tapered transition is recommended. If possible, avoid a sharp turn close to the regulator outlet and and a bull-headed tee connection to the low pressure main.
7. Install initial and delivery pressure gages to indicate performance. If the pressure rating of the delivery system or connected equipment is less than the initial steam pressure, provide a safety valve.

B. INSTALLING MAIN VALVE

1. Flush the piping system thoroughly to clear it of welding beads, scale, sand, etc.
2. Mount the main valve with diaphragm chamber down and arrow on body pointing in the direction of flow.
3. Screwed end valves should be mounted in unions.

C. INSTALLING PILOT

1. Mount the pilot on either side of the main valve by means of ¼" nipple and union provided. Make this connection to the ¼" pipe tap at the inlet of the main valve as shown in Fig. 2.
2. Screw No. 4A bleedport fitting into the ⅛" pipe tap at the outlet of the main valve body. Note bleed orifice in this fitting — vital to operation of regulator.
3. Screw No. 8B tee into ⅛" pipe tap in pilot. Select tap facing downstream.
4. Screw No. 5A elbow containing restriction orifice into ⅛" pipe tap on underside of main valve diaphragm chamber. If the initial pressure or pressure drop is less than 15 psi a No. 5B open elbow is used.
5. Connect tubing bends as illustrated in Fig. 2. Valves with condensation chamber are fitted up according to Fig. 3.

D. CONTROL PIPE

1. Use ¼" pipe for this line which connects the pilot diaphragm chamber to the desired point of pressure control. See Fig. 1.
2. Take the control at a point of minimum turbulence. Avoid control immediately at the valve outlet or after a turn. When the delivery pipe expands in size select a spot at least 4 pipe diameters beyond the point of enlargement.
3. Pitch away from pilot to avoid erratic operation and excessive fouling. Eliminate water pockets.
4. Locate delivery pressure gage in control pipe to show pressure actually reaching pilot diaphragm.

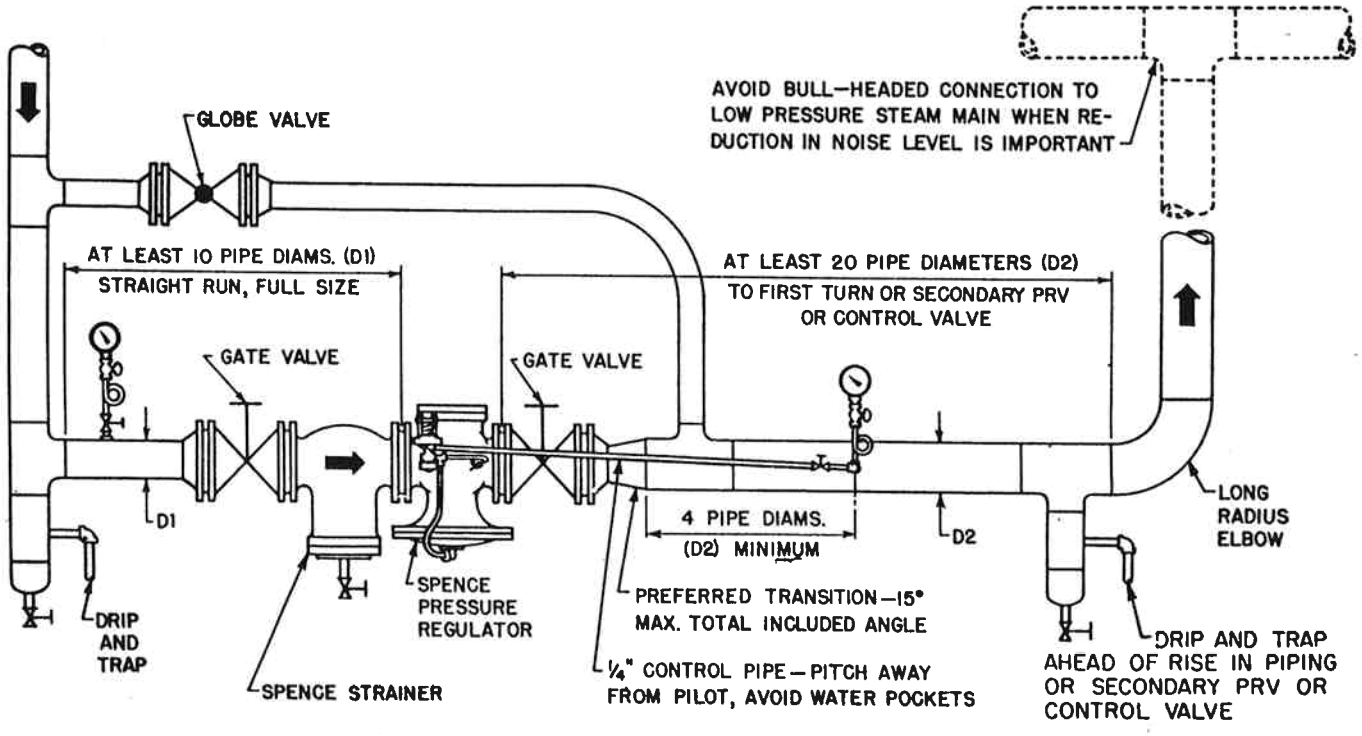


Fig. 1 — Recommended Installation of Regulator with Strainer

E. INSULATION

1. Insulation may be applied to the upper portion (globe and flanges) of the main valve. Do not insulate the diaphragm chamber, condensation chamber (if used) or any part of the pilot. See Figs. 2 and 3.

F. STARTING UP AND SETTING

1. Use by-pass to fill the delivery system and raise pressure to slightly below normal required.
2. Close pilot by releasing compression on adjusting spring. See Fig. 2.
3. Open ¼" control pipe valve. Crack outlet stop valve.
4. Crack inlet stop valve. Blow down strainer. *Caution: Never open a reducing valve without positive indication the high side is clear of condensate.*
5. Open inlet stop valve and gradually compress adjusting spring until the valve opens and takes control at desired pressure.
6. Alternately choke down on the by-pass and open outlet stop valve until the regulator is on the line.

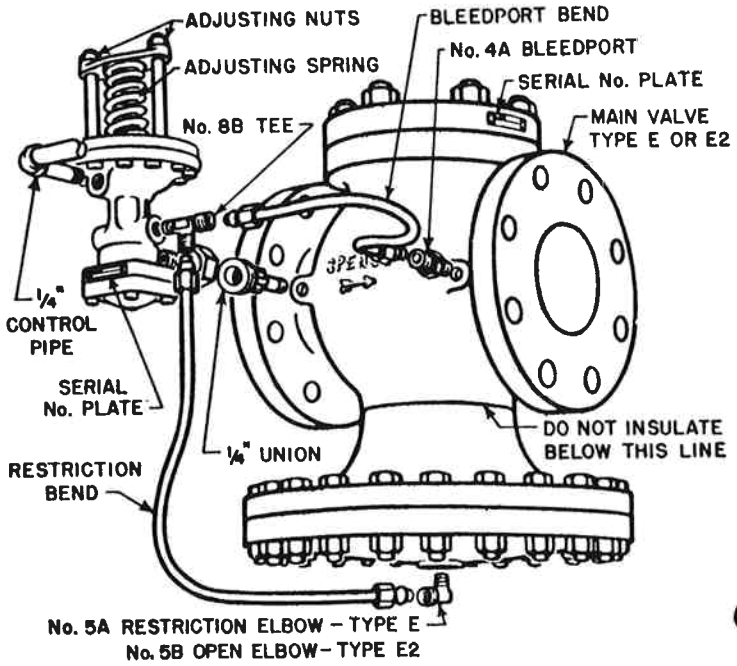


Fig. 2 — Regulator Assembly for Type E or E2 Main Valves

G. OPERATING PRINCIPLE

1. The regulator is operated by initial steam or other fluid pressure. It is normally closed, being held so by initial pressure on the disc and by an internal main spring. The pilot is opened by compressing its adjusting spring.
2. When steam is turned on, it flows through the pilot, see Fig. 2, to the No. 8B tee. Bleedport No. 4A restricts the flow, builds pressure under the diaphragm and opens the main valve. Restriction No. 5A steadies the operation of the regulator.
3. Delivery pressure feeds back through the control pipe to the pilot diaphragm. As this pressure approaches a balance with the thrust of the adjusting spring the pilot throttles the loading pressure. In turn, the main valve takes a position established by the loading pressure where just enough steam flows to maintain the set delivery pressure.

H. TROUBLE SHOOTING

1. Failure to open or sagging delivery pressure.
   a. Adjusting spring, Fig. 2, may have been tampered with.
   b. Initial pressure may be down due to partially closed supply valve, clogged strainer, or other obstruction.
   c. Orifice in No. 5A restriction elbow, Fig. 2, may be plugged. No. 4A bleedport fitting, Fig. 2, may have been omitted and an open coupling substituted.
   d. Control pipe may be plugged. Most likely points of obstruction are at shutoff valve and entrance to delivery main.
   e. Main diaphragm (20), Fig. 5, may be broken. Test with air or water before dismantling. See J2b for method of applying pressure.
2. Failure to close or over-riding delivery pressure:
   a. Adjusting spring, Fig. 2, may have been tampered with.
   b. Orifice in bleedport No. 4A may be plugged.
   c. By-pass valve may be leaking.
   d. Dust sifted on to pilot pressure plate (10), Fig. 4, may prevent closure. See J1b3.
   e. Main valve or pilot may be held open by foreign matter in seat. To determine which valve leaks:
      1. Close stop valves and ¼" control pipe valve.
      2. Remove bleedport bend, Fig. 2, so pilot will exhaust to atmosphere.
      3. Crack inlet stop valve. Steam will issue from No. 8B tee. Release compression on adjusting spring to see if pilot closes tight. Open and close several times to wash seat.
      4. Steam blowing back from bleedport means main valve disc is held open by foreign matter.
      5. Steam may wash the obstruction from the seat if the valve is made to open wide. This can be accomplished, even at light loads, if the control point is beyond the outlet stop valve as shown in Fig. 1. Reassemble bleedport bend and place regulator in operation. Then, slowly open and close outlet stop valve.

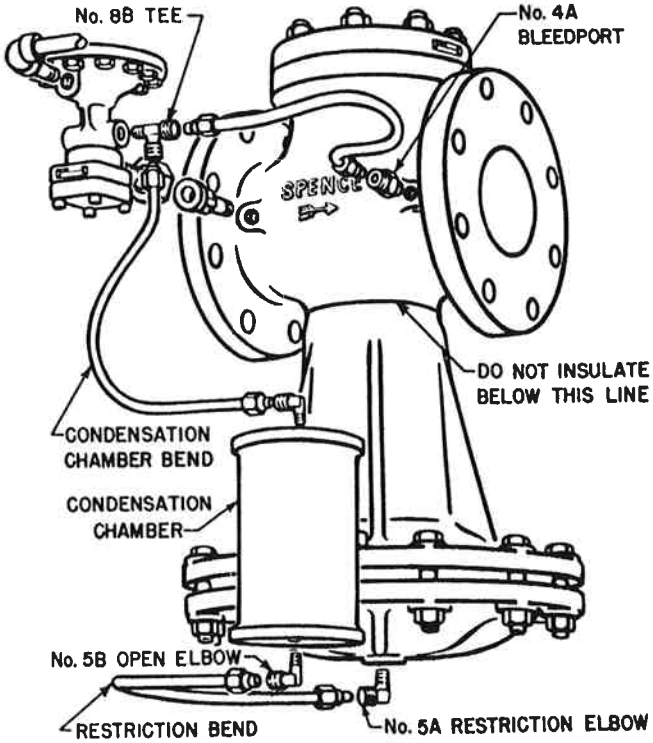


Fig. 3 — Regulator Assembly for Type E3 Main Valve with Condensation Chamber

6. Leakage of either valve requires dismantling to correct. See J2.

3. Erratic regulation may be caused by:
   a. Partial clogging of bleedport No. 4A.
   b. Water pocket in control pipe.
   c. Excessive turbulence at control point. See D2.
   d. Sticking or binding of pilot stem (16), Fig. 4. Look for deposits of compound or bits of scale on it and in bushing (15).

J. MAINTENANCE

1. Inspection
   a. Under normal conditions complete dismantling at regular intervals is not recommended. A valve kept relatively free of dirt will function for years with minimum attention.
   b. After the first few days of operation and twice a a year:
      1. Inspect for dirt collected at bleedport No. 4A and restriction elbow No. 5A, see Fig. 2.
      2. Inspect all joints for leakage. Keep bolts tight. Never allow a leak to persist.
      3. Check for dust or other air borne material which may have sifted onto the upper face of pilot pressure plate (10), Fig. 4. An accumulation here can obstruct closure of the pilot. In a very dusty atmosphere an enclosed spring chamber should be used.

2. Dismantling
   a. Pilot — Fig. 4
      1. Remove compression on spring (4).
      2. Remove diaphragm bolts (12) and take off cowl (6) and diaphragm (7).



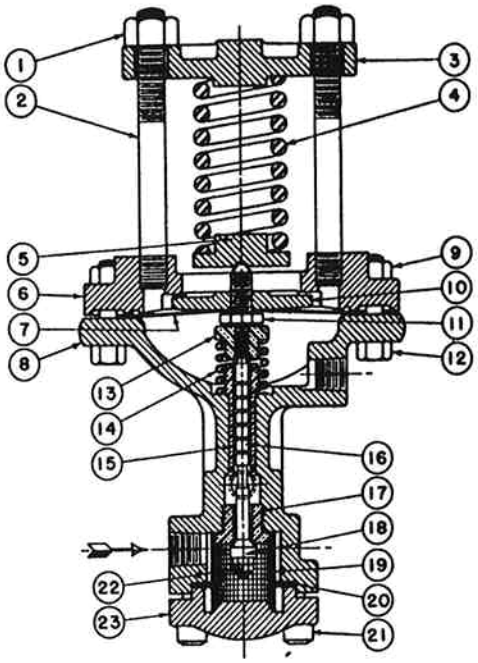
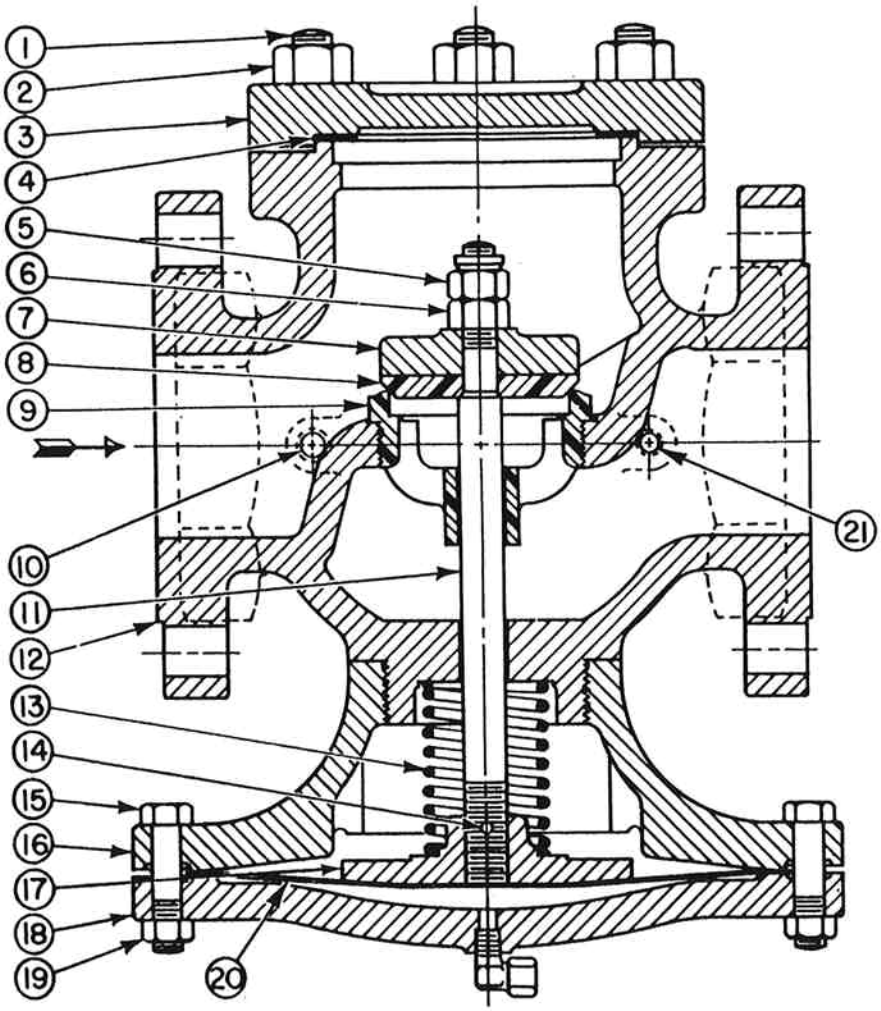


Fig. 4 — Type D Pilot



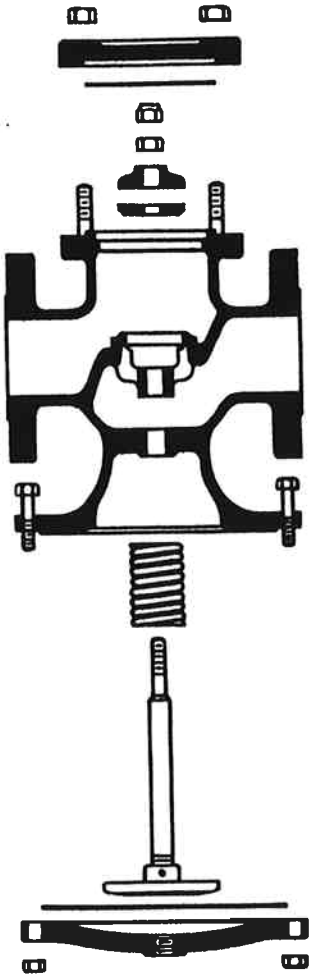


Fig. 5 — Type E Main Valve

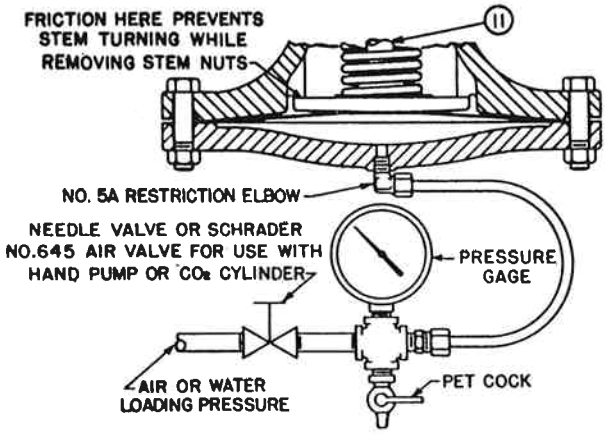


Fig. 6 — Air Loading Arrangement for Removing Stem Nuts

3. Hold pusher plate (13) with a socket wrench while turning off stem nuts (22).

b. Main Valve — Fig. 5

1. Connect a source of air or water pressure which can be adjusted by hand to the No. 5A restriction elbow. See Fig. 6.
2. Apply pressure to jack valve open and prevent stem (11) from turning while removing stem nuts (5 and 6). Usually 50 to 60 psi will suffice. Use paraffin, candle wax or penetrating oil on the threads.

c. Replacement of seat rings, main valve or pilot.

1. These joints should be made up with Copaltite, Permatex or equal plastic gasket compound.
2. Remove old compound from body and seat ring with a wire brush. Apply new compound sparingly to both parts, threads and shoulders. Let stand until tacky before assembling.

K. GRINDING IN

1. Seats and discs should never require more than the lightest touch up with very fine (400 grit) grinding compound. Heavy grinding will produce galling, wider seating surface and a groove in the disc, all of which tend to cause leakage. Reface a damaged surface before attempting to grind it in. Grind sparingly.
2. Pilot and main stems (16) and (11) are slotted for rotation with a screwdriver. Valve spring (14) or (13) is omitted from the assembly during grinding.
3. Slip the stem into its normal position. Apply compound to the disc, place it on the stem and tighten with one stem nut.
4. After grinding disassemble and clean all parts with carbon tetrachloride or kerosene.

L. VALVE SETTING

1. Pilot — Fig. 7

a. Valve setting is gaged at "C" to establish correct stem length and diaphragm position. Dimension "C" = $1\frac{1}{64}$".

b. Valve travel "A" = $\frac{3}{64}$". This movement should be checked roughly but need not be exact.

c. To install new Stem (16) omit Spring (14) and assemble other parts shown.

d. Hold Valve Disc (18) tight against its seat and screw Pusher Plate (13) down until dimension "C" = $1\frac{1}{64}$".

e. Grind off stem projection "B" flush with upper face of Pusher Plate (13)

f. Check dimension "C" and if correct lock the position by prick punching the thread at several points.

g. Caution: Burrs raised by prick punching should be scraped away. Upper face of Pusher Plate must be smooth and flat.

2. Main Valve — Fig. 8

a. Valve setting is gaged at K to establish correct stem length and diaphragm position. Dimension K is supplied with each replacement stem. For metal diaphragm valves, K is cast on the upper face of pressure plate (17).

b. To install new stem (11) grind in bevel joint with disc (8) and fasten disc firmly on stem with stem nut.

c. Insert stem and disc assembly in valve and screw on pressure plate (17). Omit spring (13) for this operation.

d. Hold disc (8) on seat and adjust position of pressure plate (17) until valve setting K is reached.

e. Push pressure plate (17) against stops in base (16).

f. Remove disc (8), drop out pressure plate and stem, drill and insert dowel pin (14) to lock the joint.

g. Grind off stem projection flush with face of pressure plate (17).

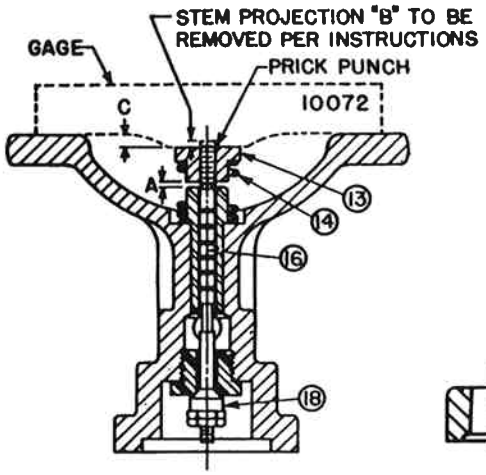


Fig. 7 — Valve Setting, Type D Pilot

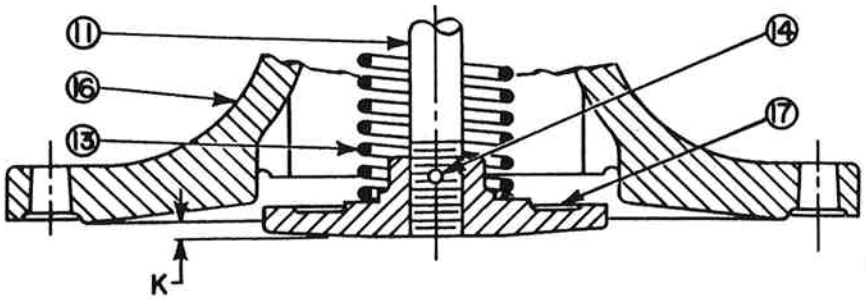


Fig. 8 — Valve Setting, Main Valve